TaxTrack

Software Requirements Specification

Project Title: TaxTrack — Transparent Tax Tracking System

Document Type	Software Requirements Specification (SRS)
Project Name	TaxTrack — Transparent Tax Tracking System
Version	1.0
Technology Stack	MERN (MongoDB, Express.js, React.js, Node.js)
Portals	Taxpayer Portal + Government Portal
Date	June 2025

1. Introduction

1.1 Purpose

The purpose of this document is to define the software requirements for TaxTrack — a MERN-stack web application that enables citizens to pay taxes online, receive unique token codes for each transaction, and transparently track how their tax money flows into government pools and is allocated to public projects.

The system features two portals: a **Taxpayer Portal** for citizens and a **Government Portal** for authorized government officials. This SRS document serves as the foundation for design, development, testing, and deployment of the system.

1.2 Scope

TaxTrack is a full-stack web application built on the MERN stack (MongoDB, Express.js, React.js, Node.js). The system enables:

- Taxpayers to register, login, and pay various types of taxes (Income Tax, GST, Property Tax, Corporate Tax, Customs Duty)
- Automatic generation of a unique token code (e.g., TXN-XXXXXXXX) upon successful payment
- Automatic routing of payments to appropriate tax pools based on tax type
- Government officials to view total pool balances, allocate funds to named public projects, and manage multiple pools
- Taxpayers to track the complete flow of their payment — from payment → pool → project allocation

1.3 Overview

This document describes the functional requirements, non-functional requirements, user interface requirements, system interfaces, performance requirements, and security requirements of the TaxTrack system. It includes entity-relationship information, data flow descriptions, class structures, and state transitions relevant to the system.

2. General Description

2.1 System Functions

The main functions of the TaxTrack system include:

Function	Description	Portal
User Registration & Login	Taxpayers register with name, email, PAN, and password. Government officials log in with official ID.	Both
Tax Payment	Taxpayers select tax type, enter amount, and submit payment. Token is generated instantly.	Taxpayer
Token Generation	A unique cryptographic-style token (TXN-XXXXXXXX) is issued per payment for tracking.	Taxpayer
Pool Auto-Routing	Payments auto-route to the correct tax pool based on tax type.	System
Fund Tracking	Taxpayers see full fund flow: payment → pool → project allocation.	Taxpayer
Pool Management	Government creates and views tax pool balances and allocations.	Government
Fund Allocation	Government allocates funds from a pool to a named project with department details.	Government
Allocation History	Complete allocation history per pool visible to both taxpayers and government.	Both
Analytics Dashboard	Government sees total collected, allocated, and unallocated funds.	Government

2.2 User Community

Taxpayers (Citizens)

Primary users who pay taxes and track their money.

- Register and log in with PAN
- Pay taxes (Income Tax, GST, Property Tax, etc.)
- Receive unique token per payment
- View payment history with token codes
- Track fund flow: payment → pool → project allocation

Government Officials

Authorized government personnel managing public funds.

- Log in with official ID and secure password
- View all tax payments received
- Create and manage multiple tax pools
- Allocate pool funds to public projects
- Monitor fund allocation with department tracking

3. Functional Requirements

3.1 Possible Outcomes

Outcome	Description
Tax Payment Submitted	Taxpayer receives a unique token code and payment is confirmed in the system.
Pool Balance Updated	The relevant tax pool balance increases by the payment amount.
Fund Allocated	Pool balance decreases and a project allocation record is created with full details.
Tracking Displayed	Taxpayer sees the complete flow: their payment, which pool it entered, and which projects rec
Auth Success/Failure	Valid credentials grant portal access; invalid credentials show an error.

3.2 Priority-Ranked Requirements

Rank	Feature	Description
1	Tax Payment with Token Generation	Core taxpayer transaction with unique token issuance
2	Fund Pool Management	Government pool creation and balance tracking
3	Fund Allocation to Projects	Government allocates from pool to public projects
4	Payment Flow Tracking	Taxpayer sees money flow from payment → pool → project
5	User Authentication (JWT)	Secure role-based login for both portals
6	Tax Pool Overview	Taxpayers see all pools and their balances
7	Audit Trail	Complete history of all payments and allocations

3.3 Input–Output Relationships

Input	Output
Taxpayer submits tax amount + type	Unique token code (TXN-XXXXXXXX) + confirmation + pool entry
Government allocates from pool	Pool balance decreases, project record created, allocation timestamp
Taxpayer views track page	Full fund flow: payment details → pool name/balance → all project allocatio
Taxpayer login credentials	JWT token + access to taxpayer dashboard
Government official ID + password	JWT token + access to government dashboard
Government creates pool	New pool with zero balance, available for tax routing

3.4 Detailed Functional Requirements

User Authentication

- The system shall allow taxpayers to register with name, email, PAN number, and password.
- The system shall authenticate users using email and password via JWT.
- The system shall provide role-based access (taxpayer vs government).
- The system shall maintain secure sessions using JSON Web Tokens (JWT) with 7-day expiry.
- The system shall prevent unauthorized access to protected API routes.

Tax Payment (Taxpayer)

- The system shall allow taxpayers to select tax type from: Income Tax, GST, Property Tax, Corporate Tax, Customs Duty.
- The system shall allow taxpayers to enter any positive amount for payment.
- The system shall generate a unique token code in format TXN-XXXXXXXX upon successful payment.
- The system shall auto-route the payment to the correct tax pool based on tax type.
- The system shall display a real-time payment success screen with token and pool info.

Payment Tracking (Taxpayer)

- The system shall display all past payments with token codes, amounts, and dates.
- The system shall allow token-based search of payment history.
- The system shall display the full fund flow for each selected payment.
- The system shall show pool details including current balance and all project allocations.
- The system shall display each project allocation from the pool in chronological order.

Pool Management (Government)

- The system shall allow government officials to create new named tax pools.
- The system shall display current balance and total collected for each pool.
- The system shall show allocation percentage as a visual progress bar.
- The system shall list all allocations per pool with project name, department, and amount.

Fund Allocation (Government)

- The system shall allow government to allocate funds from a pool to a named project.
- The system shall require project name, amount, and department for each allocation.
- The system shall prevent allocation if amount exceeds available pool balance.
- The system shall record allocation timestamp and allocating official.
- The system shall update pool balance immediately upon successful allocation.

4. User Interface Requirements

4.1 Software Interfaces

Interface	Technology	Purpose
Web Browser	Chrome, Firefox, Edge, Safari	Primary client access point
Frontend Framework	React.js 18 + Vite	Single-page application rendering
Backend API	Node.js + Express.js	RESTful API serving JSON data
Database	MongoDB + Mongoose ODM	Document-based data storage
Authentication	JWT (jsonwebtoken)	Stateless auth tokens
Password Security	bcryptjs	Password hashing and comparison
CORS	cors middleware	Cross-origin request handling

4.2 Portal Interface Examples

Taxpayer Portal Pages

- Landing Page — Portal selection with Taxpayer and Government entry points
- Login/Register Page — Toggle between login and registration with PAN input
- Dashboard — Stats cards (total paid, payments count, active pools), recent payments, pool overview, fund flow explainer
- Pay Tax Page — Tax type selector, amount input, live payment preview, success screen with token card
- Track Payment Page — Payment list with token search, detailed fund flow panel per payment

Government Portal Pages

- Login Page — Official ID and secure password with auth indicator
- Dashboard — 4 stat cards (total collected, in pools, allocated, taxpayer count), tabbed interface
- Tax Pools Tab — Pool cards with balance, progress bar, allocation list, and "Allocate" button
- All Payments Tab — Table of all payments with token, taxpayer, amount, type, pool, date
- Allocations Tab — Master list of all project allocations across all pools
- Create Pool Modal — Form to create new tax pool with name, category, description
- Allocate Funds Modal — Project name, amount, department selector, description

5. Performance Requirements

5.1 Response Time

The system should respond to user actions within the following timeframes:

Operation	Target Time	Notes
Page load / navigation	< 2 seconds	React SPA client-side routing
User login / JWT issue	< 1 second	Database lookup + token generation
Tax payment processing	< 2 seconds	DB write + token generation + pool update
Payment history fetch	< 1.5 seconds	MongoDB indexed query
Pool balance update	< 1 second	Atomic document update
Fund allocation	< 2 seconds	Pool document update + subdocument push
Dashboard analytics	< 3 seconds	Aggregation across collections

5.2 Throughput and Scalability

- The system shall handle at least 500 concurrent taxpayer sessions.
- MongoDB horizontal scaling via sharding supports future data growth.
- JWT stateless auth eliminates server-side session storage bottlenecks.
- React SPA architecture reduces server load via client-side rendering.
- Additional tax pools can be created without schema changes.

6. Non-Functional Attributes

Usability

- Both portals shall have a clear, color-coded design (Gold = Taxpayer, Green = Government).
- Tax type selection uses visual button toggles, not dropdowns.
- Payment success page uses large token display for easy reading and copying.
- Fund flow tracking uses a step-by-step visual flow (arrows and icons).
- Government dashboard uses tabbed layout to separate concerns.

Security

- All passwords stored using bcryptjs with salt rounds of 10.
- JWT tokens expire after 7 days; stored in localStorage.
- API routes protected by auth middleware checking token validity.

- Government-only routes additionally check user.role === "government".
- Token codes generated using cryptographically random alphanumeric characters.
- CORS configured to allow only specified frontend origins.

Reliability

- MongoDB ACID guarantees at the document level ensure payment atomicity.
- Pool balance and totalCollected are updated in the same Mongoose save() call.
- Token uniqueness enforced via MongoDB unique index with retry loop.
- Allocation amount validated server-side against live pool balance.

Maintainability

- Modular codebase: separate models, routes, and middleware.
- RESTful API design makes frontend/backend independently replaceable.
- Mongoose schemas with validation centralize data integrity rules.
- Environment variables via .env file for easy configuration changes.

7. Entity-Relationship Design

7.1 Entities and Attributes

Entity: USER

Attribute	Data Type	Constraint
_id	ObjectId	Primary Key
name	String	NOT NULL
email	String	UNIQUE, NOT NULL
password	String (hashed)	NOT NULL
pan	String	UNIQUE, sparse (taxpayer only)
role	Enum: taxpayer / government	DEFAULT: taxpayer
officialId	String	Government officials only
createdAt	DateTime	Auto-generated

Entity: PAYMENT

Attribute	Data Type	Constraint
_id	ObjectId	Primary Key
taxpayer	ObjectId → USER	Foreign Key
taxpayerName	String	Denormalized for query speed
amount	Number	NOT NULL, must be > 0
taxType	Enum	income_tax / gst / property_tax / corporate_tax / customs_d
description	String	Optional
tokenCode	String	UNIQUE, TXN-XXXXXXXX format
pool	ObjectId → TAXPOOL	Foreign Key
status	Enum	pending / confirmed / failed
createdAt	DateTime	Auto-generated

Entity: TAXPOOL

Attribute	Data Type	Constraint
_id	ObjectId	Primary Key

name	String	Pool display name
taxCategory	String	Tax type served by this pool
description	String	Optional
balance	Number	Current unallocated balance
totalCollected	Number	Cumulative sum of all payments
allocations	Array of ALLOCATION subdoc	Embedded subdocuments
createdBy	ObjectId → USER	Government official FK
createdAt	DateTime	Auto-generated

Entity: ALLOCATION (Subdocument of TAXPOOL)

Attribute	Data Type	Constraint
projectName	String	Name of the funded project
amount	Number	Amount allocated
department	String	Ministry/department
description	String	Optional project description
allocatedBy	ObjectId → USER	Foreign Key to Gov official
allocatedAt	DateTime	Default: Date.now

7.2 Relationships

Relationship	Entity 1	Entity 2	Cardinality	Notes
Pays	USER	PAYMENT	1 → Many	One taxpayer makes many payments
Routes To	PAYMENT	TAXPOOL	Many → 1	Many payments enter one pool
Contains	TAXPOOL	ALLOCATION	1 → Many	Embedded subdocuments
Manages	USER (Gov)	TAXPOOL	1 → Many	Government creates pools
Authorizes	USER (Gov)	ALLOCATION	1 → Many	Official allocates funds

8. API Reference

Authentication Routes

Base: /api/auth

Method	Endpoint	Auth	Description
POST	/taxpayer/register	None	Register new taxpayer with name, email, PAN, password
POST	/taxpayer/login	None	Taxpayer login — returns JWT + user object
POST	/gov/login	None	Government login with officialId + password

Payment Routes

Base: /api/payments

Method	Endpoint	Auth	Description
POST	/	Taxpayer JWT	Create tax payment, returns payment + token + pool
GET	/my	Taxpayer JWT	Get authenticated taxpayer's payment history
GET	/all	Government JWT	Get all payments across all taxpayers

Pool Routes

Base: /api/pools

Method	Endpoint	Auth	Description
GET	/	Any JWT	Get all tax pools with balances and allocations
POST	/	Government JWT	Create a new tax pool
POST	/:id/allocate	Government JWT	Allocate funds from pool to a project

9. Class Diagram

The TaxTrack system is modeled using the following classes:

Class	Attributes	Methods
User	• userId: ObjectId • name: String • email: String • password: String (hashed) • pan: String • role: Enum • officialId: String	• register() • login() • logout() • comparePassword() • getProfile()
Payment	• paymentId: ObjectId • taxpayer: Ref<User> • amount: Number • taxType: Enum • tokenCode: String • pool: Ref<TaxPool> • status: Enum • createdAt: Date	• createPayment() • generateToken() • getByTaxpayer() • getAll()
TaxPool	• poolId: ObjectId • name: String • taxCategory: String • balance: Number • totalCollected: Number • allocations: Allocation[] • createdBy: Ref<User>	• create() • addFunds(amount) • allocate(project) • getAll() • getBalance()
Allocation (Subdocument)	• projectName: String • amount: Number • department: String • description: String • allocatedBy: Ref<User> • allocatedAt: Date	• create() • getSummary()

Class Relationships

Relationship Type	Between	Description
Association	User → Payment	Taxpayer makes payments (1 to many)
Association	Payment → TaxPool	Payment routes to pool (many to 1)
Composition	TaxPool → Allocation	Pool owns allocations as embedded docs
Dependency	TaxPool ← User (Gov)	Government manages pool lifecycle
Aggregation	TaxPool ← Payment[]	Pool aggregates multiple payments

10. System Flowchart

10.1 Taxpayer Flow

Step	Activity	Decision/Output
1	User visits TaxTrack	Sees portal selection (Taxpayer / Government)
2	Selects Taxpayer Portal	Navigates to /taxpayer/login
3	Already registered?	YES → Login | NO → Register
4	Register (if new)	Enter name, email, PAN, password → Account created
5	Login	Enter email + password → JWT issued
6	Credentials valid?	YES → Dashboard | NO → Error message
7	View Dashboard	See total paid, payments count, pool overview
8	Click "Pay Tax"	Navigates to payment form
9	Select tax type + enter amount	Live preview shown
10	Submit payment	Backend creates payment, routes to pool
11	Receive token	TXN-XXXXXXXX displayed on success screen
12	Click "Track Money"	Navigates to /taxpayer/track
13	Select payment from list	Full fund flow: payment → pool → projects shown
14	Logout / Exit	Session cleared

10.2 Government Flow

Step	Activity	Decision/Output
1	Select Government Portal	Navigates to /gov/login
2	Enter Official ID + Password	Credentials verified
3	Valid credentials?	YES → Government Dashboard | NO → Error
4	View Dashboard	Stats: Total collected, in pools, allocated, taxpayer count
5	View Tax Pools Tab	All pools with balance, progress bar, allocations
6	Pool has balance?	YES → Allocate button active | NO → Pool Empty
7	Click Allocate	Modal opens: enter project name, amount, department
8	Amount ≤ pool balance?	YES → Allocate confirmed | NO → Error
9	View All Payments	Table of all taxpayer payments across system

10	View Allocations Tab	Master list of all project allocations
11	Create New Pool	Name, category, description → New pool created

11. Data Flow Diagram (DFD)

11.1 External Entities

Entity	Role	Inputs to System	Outputs from System
Taxpayer	Primary user	Registration data, login credentials, payment amount, tax type	JWT token, token code, fund tracking data
Government Official	Admin user	Official credentials, pool data, allocation details	JWT token, pool analytics, all payment data

11.2 Data Stores

ID	Data Store	Contents	Technology
D1	User Store	Taxpayer & government accounts, hashed passwords, PAN, official ID	MongoDB Collection
D2	Payment Store	All tax payments, token codes, amounts, types, pool references	MongoDB Collection
D3	Pool Store	Tax pools, balances, totalCollected, allocation sub-documents	MongoDB Collection

11.3 Level 0 DFD — Context Diagram

The entire TaxTrack system is represented as a single process receiving inputs from Taxpayer and Government entities and returning appropriate outputs.

From Entity	Data Flow to System	System Process	Data Flow from System	To Entity
Taxpayer	Registration/login data	TaxTrack System	JWT + dashboard access	Taxpayer
Taxpayer	Payment amount + type	TaxTrack System	Token code + pool info	Taxpayer
Taxpayer	Track request	TaxTrack System	Fund flow details	Taxpayer
Government	Official credentials	TaxTrack System	JWT + dashboard access	Government
Government	Allocation request	TaxTrack System	Updated pool + allocation record	Government

11.4 Level 1 DFD — Process Decomposition

Process ID	Process Name	Inputs	Outputs	Data Stores Used
P1.0	User Auth	Credentials, registration data	JWT token, user object	D1 – Users
P2.0	Tax Payment	Amount, tax type, user JWT	Token code, payment record, pool update	D2 – Payments, D3 – Pools
P3.0	Pool Routing	Tax type	Selected pool ID	D3 – Pools
P4.0	Fund Tracking	User JWT	Payment history + pool + allocations	D2, D3
P5.0	Pool Management	Gov JWT, pool data	Pool created/updated	D3 – Pools

P6.0	Fund Allocation	Gov JWT, project details, amount	Allocation record, pool balance update	D3, P data
P7.0	Analytics	Gov JWT	Aggregated financial stats	D2, D3

12. State Diagrams

12.1 Payment State Transitions

Current State	Event	Next State	Action
(Initial)	Taxpayer submits payment	Pending	Record created, pool routing begins
Pending	Payment DB write successful	Confirmed	Token generated, pool balance updated
Pending	DB write fails / error	Failed	Error returned to taxpayer
Confirmed	Taxpayer views tracking	Confirmed	Fund flow displayed (no state change)
Confirmed	(Final state)	—	Payment remains in confirmed state permaner

12.2 Pool State Transitions

Current State	Event	Next State	Action
(Initial)	Government creates pool	Empty (balance=0)	Pool record created
Empty	Tax payment routed to pool	Active (balance>0)	balance += amount, totalCollected += a
Active	More payments routed	Active	balance increases further
Active	Government allocates funds	Active (or Empty)	balance -= allocation amount
Active	All funds allocated	Empty (balance=0)	Pool ready for new payments

12.3 Sequence Diagram — Tax Payment Flow

Step	Actor	Message	Recipient	Response
1	Taxpayer	POST /api/payments (amount, taxType)	React Client → Express API	
2	Auth Middleware	Verify JWT token	JWT Library	User object decoded
3	Payment Controller	assignPool(taxType)	MongoDB	Pool document returned
4	Payment Controller	generateToken()	Internal	Unique TXN-XXXXXXXX
5	Payment Controller	Payment.create(...)	MongoDB	Payment document saved
6	Payment Controller	pool.balance += amount; pool.save()	MongoDB	Pool updated
7	Express API	Return {payment, pool}	React Client	—
8	React Client	Display success screen with token	Taxpayer	—

13. Decision Table

13.1 Tax Payment Decision Table

Condition / Action	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5
User Authenticated?	Yes	Yes	Yes	No	Yes
Amount > 0?	Yes	No	Yes	—	Yes
Tax Pool Available?	Yes	—	No	—	Yes
Token Unique?	Yes	—	—	—	No (retry)
— Actions —					
Create Payment Record	Yes	No	No	No	Yes (retry)
Generate Token	Yes	No	No	No	Yes (new)
Update Pool Balance	Yes	No	No	No	Yes
Return Success + Token	Yes	No	No	No	Yes
Return Error Message	No	Yes	Yes	Yes (401)	No

13.2 Fund Allocation Decision Table

Condition / Action	Rule 1	Rule 2	Rule 3	Rule 4
Official Authenticated?	Yes	Yes	No	Yes
Pool Exists?	Yes	Yes	—	No
Amount ≤ Pool Balance?	Yes	No	—	—
— Actions —				
Deduct from Pool Balance	Yes	No	No	No
Create Allocation Record	Yes	No	No	No
Return Updated Pool	Yes	No	No	No
Return Error (insufficient)	No	Yes	No	No
Return 401 Unauthorized	No	No	Yes	No
Return 404 Pool Not Found	No	No	No	Yes

14. Schedule and Budget

14.1 Development Timeline

Phase	Duration	Key Deliverables
Requirement Analysis	1 Week	SRS Document, stakeholder review
System Design	1 Week	ER Diagram, API Design, UI Wireframes
Backend Development	1.5 Weeks	Express API, MongoDB models, JWT auth, pool routing
Frontend Development	1.5 Weeks	React pages, both portals, fund flow visualization
Integration & Testing	1 Week	API integration, functional testing, security testing
Documentation	0.5 Weeks	SRS, Lab Document, README
Deployment (Optional)	0.5 Weeks	Heroku/Vercel/MongoDB Atlas setup
Total	7 Weeks	

14.2 Cost Estimate

Item	Tool/Service	Cost
Development Tools	VS Code, Postman	Free
Frontend Framework	React.js + Vite	Free (Open Source)
Backend Framework	Node.js + Express.js	Free (Open Source)
Database	MongoDB Community / Atlas Free Tier	Free
Authentication	jsonwebtoken, bcryptjs	Free (Open Source)
Hosting — Frontend	Vercel (Free tier)	$0
Hosting — Backend	Render (Free tier)	$0
Database Hosting	MongoDB Atlas M0 (Free)	$0
Estimated Total		$0 – $20 (if paid hosting used)

15. Appendices

15.1 Token Code Format

Each tax payment generates a unique token in the format: **TXN-XXXXXXXX**

Where X is a random character from: A-Z (excluding I, O) and 2-9 (to avoid visual confusion).

Example tokens: TXN-AJ7KP3MN, TXN-B5QRXZ2H, TXN-CN4MWVYA

15.2 Tax Type to Pool Mapping

Tax Type	Auto-Assigned Pool	Rationale
Income Tax	Infrastructure Development Fund	Primary source for road/rail/bridge projects
Property Tax	Infrastructure Development Fund	Urban infrastructure linkage
GST (Goods & Services)	Healthcare & Welfare Pool	Consumer-linked welfare programs
Corporate Tax	Education & Research Fund	Business-education linkage
Customs Duty	Infrastructure Development Fund	Trade infrastructure support

15.3 Glossary

Term	Definition
Token Code	Unique alphanumeric identifier (TXN-XXXXXXXX) issued for each tax payment
Tax Pool	Accumulated fund from tax payments, categorized by tax type or purpose
Allocation	Transfer of funds from a pool to a named public project
JWT	JSON Web Token — stateless authentication mechanism
PAN	Permanent Account Number — Indian taxpayer identification
MERN	MongoDB, Express.js, React.js, Node.js — the technology stack used
ODM	Object Document Mapper — Mongoose's MongoDB interface layer
Subdocument	Embedded MongoDB document within a parent document (allocations within a pool)
Role-Based Access	System restricts API access based on user.role (taxpayer vs government)